

NovaTech Data Verification Log

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Instructions

Query each pre-indexed knowledge base using Quick Chat. For each question, record the expected answer (from the data dictionary), Q's actual response, and whether they match. Minimum 6 entries (2 per data knowledge base).

Verification Log

#	Knowledge Base	Question Asked	Expected Answer	Q's Actual Answer	Match?	Notes
1	NovaTech CRM Deals	Are all values in <code>opportunity_id</code> unique, and does each record represent an isolated deal?	The data dictionary defines <code>opportunity_id</code> as a "Unique identifier for each sales opportunity" in the format <code>OPP-XXXXX</code> with None under Nulls.	No — opportunity_id values are not unique, and each record does not represent an isolated deal.	no	The data is aggregated at the company level , not at the individual opportunity level.
2	NovaTech CRM Deals	Are there any chronological date anomalies where <code>deal_closed_date</code> occurs before	Both <code>deal_created_date</code> and <code>deal_closed_date</code> are defined as valid Date (YYYY-MM-DD) fields with	No chronological date anomalies were found. The query against the underlying dataset confirmed that there are zero	yes	

		deal_created_date?	None under Nulls, and users can compute days-to-close directly as <code>deal_closed_date - deal_created_date</code> (implying standard forward chronology).	<code>records</code> where <code>deal_closed_date</code> occurs before <code>deal_created_date</code>		
3	NovaTech Marketing Campaigns	Are all records uniquely identified by <code>lead_id</code> without duplicate customer touchpoints?	Yes. The data dictionary defines <code>lead_id</code> as a "Unique identifier for each marketing lead" with None under Nulls, explicitly noting "One row per lead" across all 2,240 records	No — <code>lead_id</code> does not uniquely identify records. Every single <code>lead_id</code> in the dataset appears multiple times.	no	Every <code>lead_id</code> is duplicated — there are zero uniquely-occurring leads. This is because the dataset is a denormalized (unified) join across CRM, Marketing, and Support data sources. Each lead gets repeated for every associated opportunity and support ticket row
4	NovaTech Marketing Campaigns	Are there any invalid, negative, or corrupt monetary values in <code>campaign_spend</code> or <code>revenue_attributed</code> ?	No. <code>campaign_spend</code> is defined as a float ranging from 501.71 to 14,995.14 with None under Nulls, while <code>revenue_attributed</code> is defined as a float ranging from	No invalid, negative, or corrupt monetary values were found in either <code>campaign_spend</code> or <code>revenue_attributed</code> .	yes	

			0.00 to 5,575.51 (with 0.00 specifically reserved for non-converting leads where campaign_response = 0).			
5	NovaTech Support Tickets	Does the ticket_id column satisfy primary key uniqueness constraints across all logged issues?	By specification, ticket_id is defined as the "Unique identifier for each support ticket" in the format TKT-XXXXXX with None under Nulls	No — ticket_id does not satisfy primary key uniqueness constraints. Every single ticket_id in the dataset is duplicated.	no	The novatech_Unified_Moamen dataset is a denormalized join across three data domains — Sales (CRM), Marketing, and Support. Each ticket row fans out across every matching opportunity and lead record for the same account, creating a multiplicative effect.
6	NovaTech Support Tickets	How many tickets are currently active and unresolved, and does their presence align with missing post-ticket surveys?	Exactly 59 rows (2.0%) are expected to be null in ticket_resolved_date. The dictionary notes this field is blank for unresolved tickets, and resolution time is calculated as ticket_resolved_date -	There are 58 distinct tickets with a null ticket_resolved_date, meaning they are currently active and unresolved out of a total of 2,792 distinct tickets (2.08% of all tickets).	yes	

			ticket_created_date. customer_sentiment lists valid values as <i>negative</i> , <i>neutral</i> , <i>positive</i> with None under Nulls			
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Cross-Check

Pick one fact from above and confirm it independently in the QuickSight dataset preview.

- **Fact verified:** The count of active, unresolved support tickets (ticket_resolved_date is null) and total tickets within the joined CRM-matched dataset (novatech_Unified_Moamen)
- **Chat said:** There are 58 distinct tickets with a null ticket_resolved_date, meaning they are currently active and unresolved out of a total of 2,792 distinct tickets (2.08% of all tickets).
- **QuickSight shows:** In the joined dataset where CRM accounts are matched (ACCT-001 through ACCT-085), exactly **58 distinct tickets** (58 rows) have an empty/null ticket_resolved_date out of **2,792 distinct tickets** (2,796 total rows, excluding the 204 orphan support tickets in ACCT-101 through ACCT-115)
- **Consistent?** Yes